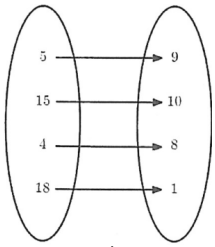
Domain, Range and Composite Functions

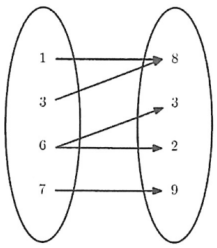
Questions 1–39

Question 1. Identify which of the following mapping diagrams depict a function:

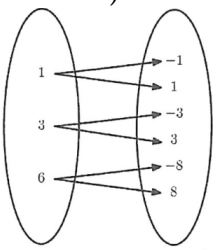
a)



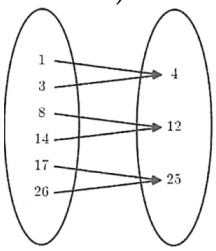
b)



c)

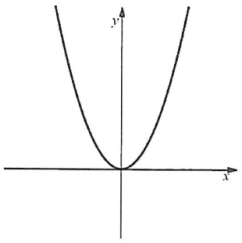


d)

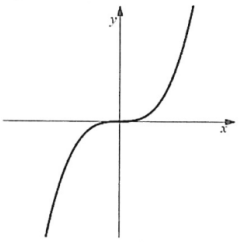


Question 2. For each of the following, state whether it is one-to-one, many-to-one, one-to-many or many-to-many and if it is function or not:

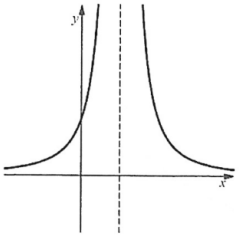
a)



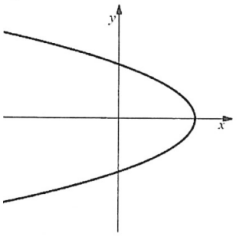
b)



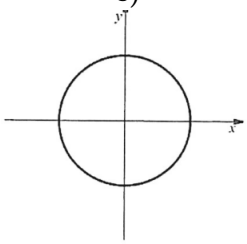
c)



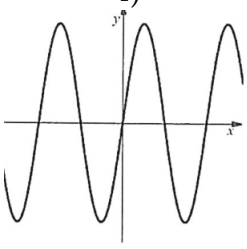
d)



e)



f)



Question 3. It is given that $f : x \mapsto 5x - 3$ where $x \in \{1, 2, 3, 4\}$.

a) Represent f on a mapping diagram.

b) State whether $f(x)$ is a function and if so, what type of function it is.

Question 4. It is given that $f(x) = 3x^2 - 1$ where $x \in \{-2, -1, 0, 1\}$.

a) Represent $f(x)$ on a mapping diagram.

b) State whether $f(x)$ is a function and if so, what type of function it is.

Question 5. It is given that $f : x \mapsto \sqrt{x-1}$ where $x \in \{3, 5, 8, 10\}$.

a) Represent $f(x)$ on a mapping diagram.

b) State whether $f(x)$ is a function and if so, what type of function it is.

Question 6. Determine which of these functions are one-to-one and which are many to one:

a) $f(x) = 4x - 3,\ \{x : x \in \mathbb{R}\}$

b) $f : x \mapsto x^2 + 5,\ \{x : x \in \mathbb{R}\}$

c) $f(x) = \frac{1}{x},\ \{x : x \in \mathbb{R}, x \neq 0\}$

d) $f : x \mapsto 2x^3,\ \{x : x \in \mathbb{R}\}$

e) $f(x) = x(x + 4),\ \{x : x \in \mathbb{R}\}$

f) $f : x \mapsto 3\cos x,\ \{x : x \in \mathbb{R}, 0° \leq x \leq 180°\}$

g) $f(x) = (x - 7)^2 + 3,\ \{x : x \in \mathbb{R}, x > 7\}$

h) $f : x \mapsto x^2(x - 4),\ \{x : x \in \mathbb{R}, x > 0\}$

i) $f(x) = 3\ln x,\ \{x : x \in \mathbb{R}, x > 0\}$

j) $f : x \mapsto \sqrt{x},\ \{x : x \in \mathbb{R}, x \geq 4\}$

Question 7. The answers for the following questions which require a range of values should be given in set notation State the range of the following, given that in each case $x \in \mathbb{R}$:

a) $f(x) = 1 + 2^{-x}$

b) $y = (x+3)^2 - 5$

c) $f(x) = x(x-2)^2$

d) $y = -2e^{-4x}$

e) $f(x) = 5 - 3^{-x}$

Question 8. For each of the following functions $y = f(x)$ below, state the largest possible domain and the corresponding range of f:

a) $y = \cos x$

b) $y = \sqrt{x}$

c) $y = -(x+4)^2 - 9$

d) $y = 2x^2 - 6x + 9$

e) $y = \frac{1}{x} + 4$

f) $y = \frac{1}{x-3}$

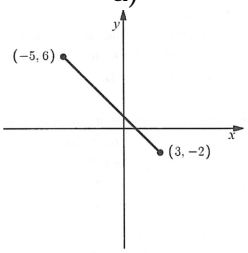
g) $y = 2 + e^x$

h) $y = 4\ln x$

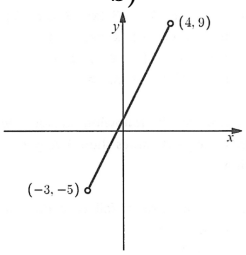
i) $y = 3\ln(x-2)$

Question 9. For each of the graphs below, state the domain and the range:

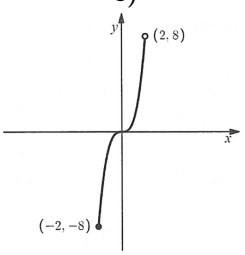
a)



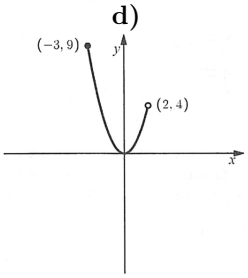
b)



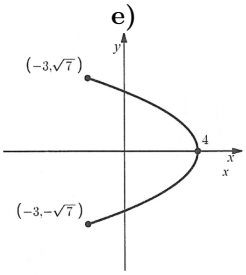
c)



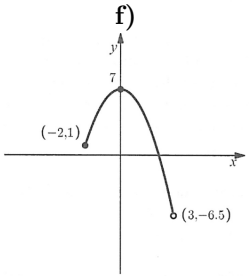
d)



e)

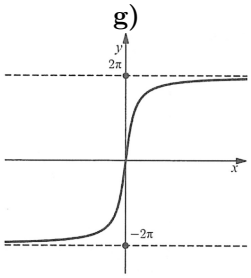


f)

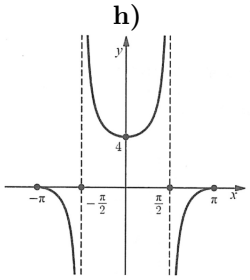


Question 9 (continued). For each of the graphs below, state the domain and the range:

g)



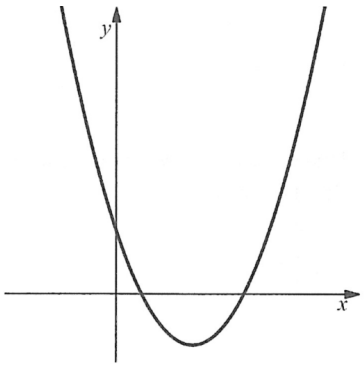
h)



Question 10. Sketch a graph which is many-to-one, has domain $\{x : x \in \mathbb{R}\}$, range $\{y : y \in \mathbb{R}, y \leq 3\}$, the y-intercept is negative, and its only maximum occurs at $x = p$ where $p > 0$.

Question 11. Sketch a graph which is one-to-one, has domain $\{x : x \in \mathbb{R}\}$, range $\{y : y \in \mathbb{R}\}$, the y-intercept is positive and it has a single stationary point which occurs at $x = p$ where $p > 0$.

Question 12. The graph below shows $f(x) = \frac{1}{2}(x-3)^2 - 2$.



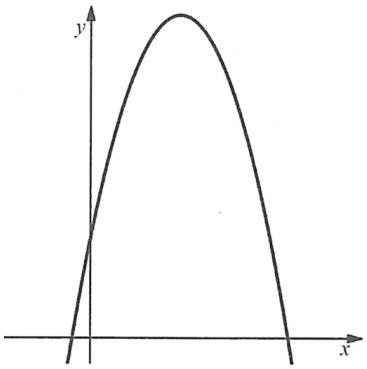
Find the range of $f(x)$ when the domain is restricted as follows:

a) $x \geq 5$

b) $x \leq 2$

c) $0 \leq x \leq 4$

Question 13. The graph below shows $f(x) = -0.5x^2 + 5x + 5.5$.



Find the range of $f(x)$ when the domain is restricted as follows:

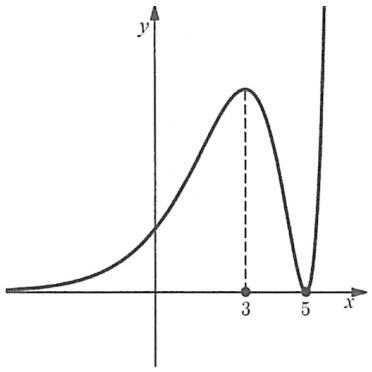
a) $x \geq 5$

b) $x \leq 2$

c) $0 < x < 4$

d) $3 \leq x \leq 7$

Question 14. The graph below shows $f(x) = \frac{1}{12}(x-5)^2 e^x$.



Find the range of $f(x)$ when the domain is restricted as follows:

a) $x < 5$

b) $3 \leq x \leq 5$

c) $x \geq 5$

d) $0 \leq x < 3$

Question 15. Given $f : x \mapsto x^2 + 8x + 4$.

a) Sketch the graph of $y = f(x)$, labelling any intersections with the axes and any stationary points.

b) State its range for $x \in \mathbb{R}$.

Question 16. Find the range of each of the following functions:

a) $f(x) = 4x + 1,\ \{x : x \in \mathbb{R}, x \geq 1\}$

b) $f : x \mapsto \frac{1}{3}x - 2,\ \{x : x \in \mathbb{R}, x < -2\}$

c) $f(x) = 5 - 8x,\ \{x : x \in \mathbb{R}, -2 \leq x < 4\}$

d) $f(x) = 2\sqrt{x},\ \{x : x \in \mathbb{R}, 4 < x \leq 16\}$

e) $f : x \mapsto 3x - x^2,\ \{x : x \in \mathbb{R}, 0 < x < 3\}$

f) $f(x) = \frac{1}{1+x^2},\ \{x : x \in \mathbb{R}\}$

Question 17. Given that for $x \in \mathbb{R}$, $f(x) = \dfrac{1}{2x^2 - 4x + 9}$. By first completing the square on $2x^2 - 4x + 9$ state the range of f.

Question 18. The function $f(x) = 2x^2 + kx + 13$, where k is a constant, has domain $x \in \mathbb{R}$. It is known that the range of f is $f(x) \geq -5$. Find the possible value(s) of k.

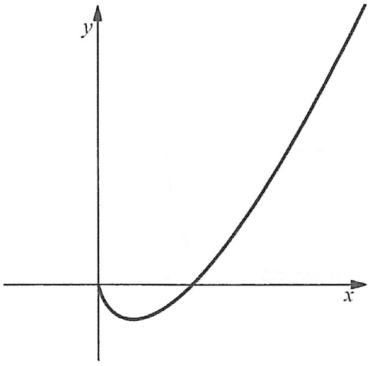
Question 19. The function $f(x) = k(x - a)(x - b)$ where a, b, k are all positive, is restricted in its domain such that it is one-to-one and $x \geq p$ where p gives the largest possible range. Find the value of p giving your answer in terms of a and b.

Question 20. Find the largest possible domain and range of

a) $f : x \mapsto -x^4 + 4x$

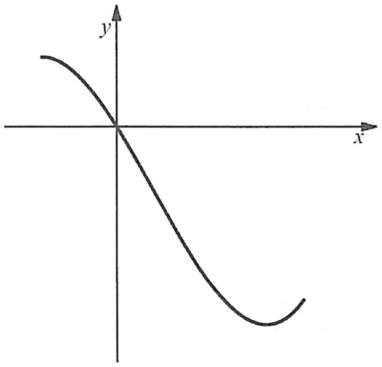
b) $f : x \mapsto 4x - e^{2x}$

Question 21. Prerequisite knowledge for Q21: Differentiation of standard functions and product rule
The graph below shows $f(x) = x \ln x$.



Find the largest possible domain and range of $f : x \mapsto x \ln x$.

Question 22. The graph below shows $f(x) = x^3 - 3x^2 - 24x$ for $-2 < x < 5$.



Find the range of

a) f

b) g given that $g(x) = 2f(x)$

c) h given that $h(x) = 3f(x) - 4$

Question 23. Given $f : x \mapsto 5x - 1$ and $g : x \mapsto x^2 + 2$, find

a) $fg(-1)$

b) $gf(2)$

c) $ff(4)$

Question 24. Given $f(x) = x^2$ and $g(x) = \dfrac{1}{x-2}$, find $fg(4)$.

Question 25. Given $f(x) = 2x + 1$ and $g(x) = x^2$, find

a) $fg(x)$

b) $gf(x)$

c) $ff(x)$

d) the value of a such that $fg(a) = 33$ and a is positive

e) the values of x such that $fg(x) = gf(x)$

Question 26. Given $f(x) = \frac{1}{x}$ find

a) $f^{89}(x)$

b) $f^{100}(x)$

Question 27. Given $f(x) = 1 + \sqrt{x}$ and $g(x) = 3 - 2^{-x}$. Solve $gf(x) = \frac{23}{8}$.

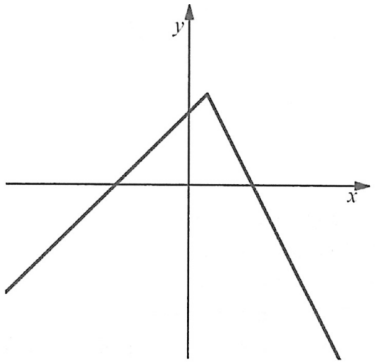
Question 28. Given $f : x \mapsto 5\log_{10} x$ and $g : x \mapsto 10^x$

a) find $fg(x)$

b) show that $fgf(x) = 25\log_{10} x$

Question 29. A function is given by $f(x) = -kx^2 + 12x - 11$ where k is a positive constant. It is known that $ff(2) = -2$. Find the possible values of k.

Question 30. The graph below shows $f(x) = \begin{cases} x+4, & x \leq 1 \\ 7-2x, & x > 1. \end{cases}$

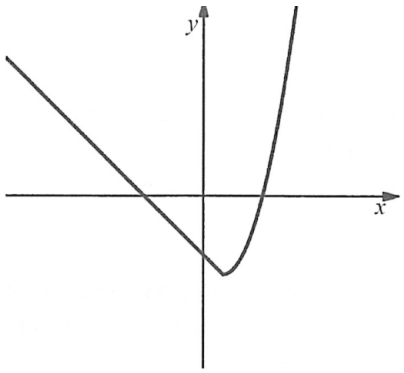


a) State the value of $ff(1)$

b) Find the value of $ff(\frac{1}{2})$

c) Find the range of values of x for which $f(x) < -2$

Question 31. The graph below shows $f(x) = \begin{cases} -x-3, & x < 1 \\ x^2-2x-3, & x \geq 1. \end{cases}$



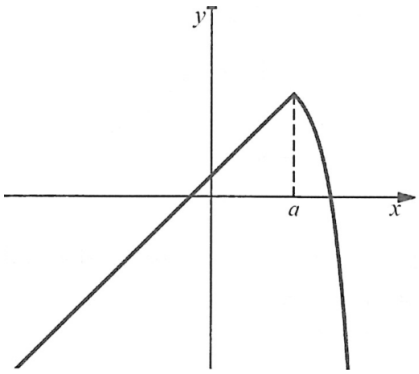
a) State the value of $ff(1)$

b) State the range of f

c) Find the value of $ff(2)$

d) Find the range of values of x for which $f(x) < 0$

Question 32. The graph below shows $f(x) = \begin{cases} x+1, & x < a \\ 6 - e^{x-4}, & x \geq a. \end{cases}$



The range of f is $f(x) \leq 5$. Find

a) the value of a

b) the exact value of $ff(\frac{7}{2})$

c) the range of values of x for which $f(x) \geq 0$

Question 33. Given $f : x \mapsto \dfrac{2x+1}{x-5}$, $x \neq 5$

a) Find the range of f

b) Find $ff(x)$

Question 34. Given $f(x) = \dfrac{ax+1}{x-4}$, $x \neq 4$ where a is a constant and $a \neq -\frac{1}{4}$

a) Find the range of f

b) Find $ff(x)$

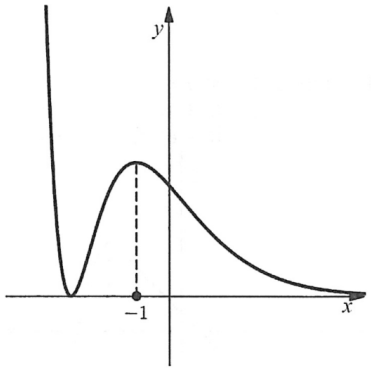
Question 35. Given $f(x) = x^2 - ax$ and $g(x) = a^2 - x$ where a is a positive constant and $fg(2) = 0$

a) find $fg(x)$

b) find the possible values of a giving a in exact form.

Question 36. The answers for the following questions which require a range of values should be given in interval notation

The graph below shows $f(x) = 2(x+3)^2 e^{-x}$.



Find the range of $f(x)$ when the domain is restricted as follows:

a) $x < -3$

b) $-3 \leq x \leq -1$

c) $x \geq 0$

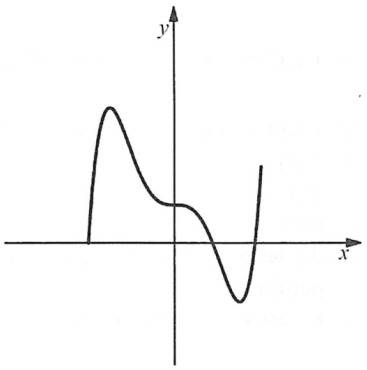
d) $-3 \leq x < 0$

Question 37. Given $f(x) = 2x^2 - 12x + 13$

a) Sketch the graph of $y = f(x)$, labelling any intersections with the axes and any stationary points.

b) State its range for $x \in \mathbb{R}$.

Question 38. The graph below shows $f(x) = 16x^5 - 15x^3 + 1$ for $-1 < x < 1$.



Find the range of

a) f

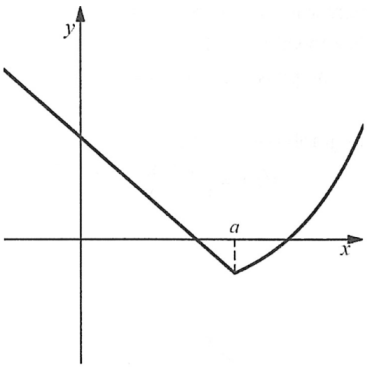
b) g given that $g(x) = 64f(x)$

c) h given that $h(x) = 8f(x) - 10$

Find the domain of

d) k given that $k(x) = f(2x)$

Question 39. The graph below shows $f(x) = \begin{cases} 3 - kx, & x < a \\ e^{x-2} - 2, & x \geq a \end{cases}$ where k is a positive constant.



The range of f is $f(x) \geq -1$.

a) Find the value of a

b) Find the value of k

c) Find the exact value of $ff(3)$

d) Find the range of values of x for which $f(x) \leq 0$